TWISTS RESTRICT TO TRIVIAL INVERTIBLE SHEAVES ON STANDARD OPENS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Twist sheaves and multiplication maps. chapter definitions of the twists and the multiplication maps (section-level, not carded). O_X(nd) and M(nd)-tilde are the tildes of the shifted graded modules, and the three multiplication maps are induced by the shift identifications.

Source reference. Original equation 01MO: sources/stacks-constructions/pinned-original-source.tex, source lines 1930–1930. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 01MQ: sources/stacks-constructions/pinned-original-source.tex, source lines 1966–1966. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 03GJ: sources/stacks-constructions/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01CS: focused-seeds/geometry/S9-cons-when-invertible/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. Set $X = Proj(S)$. Let $f \in S$ be homogeneous of degree $d > 0$. The sheaves $\mathcal{O}_X(nd)|_{D_+(f)}$ are invertible, and in fact trivial for all $n \in \mathbf{Z}$ (see Modules, Definition 01CS). The maps (01MO) restricted to $D_+(f)$*

$$\mathcal{O}_X(nd)|_{D_+(f)} \otimes_{\mathcal{O}_{D_+(f)}} \mathcal{O}_X(m)|_{D_+(f)} \longrightarrow \mathcal{O}_X(nd+m)|_{D_+(f)},$$

the maps (03GJ) restricted to $D_+(f)$

$$\mathcal{O}_X(nd)|_{D_+(f)} \otimes_{\mathcal{O}_{D_+(f)}} \mathcal{F}(m)|_{D_+(f)} \longrightarrow \mathcal{F}(nd+m)|_{D_+(f)},$$

and the maps (01MQ) restricted to $D_+(f)$

$$\widetilde{M}(nd)|_{D_+(f)} = \widetilde{M}|_{D_+(f)} \otimes_{\mathcal{O}_{D_+(f)}} \mathcal{O}_X(nd)|_{D_+(f)} \longrightarrow \widetilde{M(nd)}|_{D_+(f)}$$

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are isomorphisms for all $n, m \in \mathbf{Z}$.

Proof. The graded S-module maps $S \to S(nd)$, and $M \to M(nd)$, given by $x \mapsto f^n x$ become isomorphisms after inverting f. The first shows that $S_{(f)} \cong S(nd)_{(f)}$ which gives an isomorphism $\mathcal{O}_{D_+(f)} \cong \mathcal{O}_X(nd)|_{D_+(f)}$. The second shows that the map $S(nd)_{(f)} \otimes_{S_{(f)}} M_{(f)} \to M(nd)_{(f)}$ is an isomorphism. The case of the map (03GJ) is a consequence of the case of the map (01MO). $\square$